概率论. 矩母函数与特征函数专题

一. 常用矩母函数.

① $X \sim B(n,p)$

$M(t)=E(e^{tX})=(q+pe^t)^n$

② $X \sim G(p)$

$M(t)=E(e^{tX})=\frac{pe^t}{1-qe^t}$

③ $X \sim Pasca(r,p)$

$M(t)=E(e^{tX})=\left(\frac{pe^t}{1-qe^t}\right)^r$

④ $X \sim P(\lambda)$

$M(t)=e^{\lambda(e^t-1)}$

⑤ $X \sim U(a,b)$

$M(t)=\frac{e^{tb}-e^{ta}}{t(b-a)}$.

⑥ $X \sim E(\lambda)$

$M(t)=\frac{\lambda}{\lambda-t} \quad t<\lambda$

⑦ $X \sim \Gamma(\alpha,\lambda)$.

$M(t)=\left(\frac{\lambda}{\lambda-t}\right)^\alpha \quad t<\lambda$.

proof: $\Gamma(\alpha)=\int_0^{+\infty} x^{\alpha-1}e^{-x}dx$

$M(t)=E(e^{tx})=\int_0^{+\infty}\frac{e^{tx}\,\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}dx$

$=\int_0^{+\infty}\frac{\lambda^\alpha}{\Gamma(\alpha)}e^{(t-\lambda)x}x^{\alpha-1}dx \quad t<\lambda$

$=\left(\frac{\lambda}{\lambda-t}\right)^\alpha$.

⑧ 设 $X \sim N(\mu,\sigma^2)$

$M(t)=e^{\mu t+\frac{1}{2}\sigma^2t^2}$

特征函数: 即 $e^t \longrightarrow e^{it}$.

特例 1: $U(-b,b) \Rightarrow \phi(t)=\frac{e^{ibt}-e^{-ibt}}{it(2b)}$

$=\frac{2i\sin bt}{2ibt}=\frac{\sin bt}{bt}$

$b=\inf \quad \phi(t)=\frac{\sin t}{t}$

$N(\mu,\sigma^2) \Rightarrow \phi(t)=e^{\mu it-\frac{1}{2}\sigma^2t^2}$

$\sigma=1,\ \mu=0 \quad \phi(t)=e^{-\frac{1}{2}t^2}$.

$M(t)=e^{\frac{1}{2}t^2}$.

二、基本应用

应用一：生成矩。

$M^{(k)}(0) = E(X^k)$

应用二：独立性 $\xRightarrow{\text{证明}}$ 可加性

利用 $M_{X+Y}(t) \overset{X\perp Y}{=} M_X(t) M_Y(t)$

例：$X \sim U(0,b) \Rightarrow M(t) = \dfrac{e^{tb}-e^{ta}}{t(b-a)}$

$X \sim U(0,1)$ 时 $M(t) = \dfrac{e^t-1}{t} \overset{\text{Taylor}}{=\!=} \dfrac{\sum_{k=0}^{+\infty} \frac{t^k}{k!}}{t} = \sum_{k=1}^{+\infty} \frac{t^{k-1}}{k!}$

$$= \sum_{k=0}^{+\infty} \frac{t^k}{(k+1)!}$$

$$\Rightarrow M^{(k)}(0) = \frac{k!}{(k+1)!} = \frac{1}{k+1} = E(X^k)$$

例：$X \sim N(0,1)$ $M(t) = e^{\frac{1}{2}t^2} \overset{\text{Taylor}}{=\!=} \sum_{k=0}^{+\infty} \frac{\left(\frac{1}{2}t^2\right)^k}{k!} = \sum_{k=0}^{+\infty} \frac{t^{2k}}{2^k k!}$

$$E(X^{2k-1}) = 0$$

$E(X^{2k}) \Rightarrow$ 对比系数：$\dfrac{E(X^{2k})}{(2k)!} = \dfrac{1}{2^k k!}$

$$\Rightarrow E(X^{2k}) = \frac{(2k)!}{2^k k!} = (2k-1)!!$$

三. 联合矩母函数的应用

$X_1 \cdots X_n \quad M(t_1 \cdots t_n) = E\left(e^{t_1 X_1 + \cdots + t_n X_n} \right)$

$= E\left(e^{t_1 X_1} e^{t_2 X_2} \cdots e^{t_n X_n} \right)$

$t_1 = 0$ 时的形式 $\quad M_{X_1, X_2}(t_1, t_2) = E\left(e^{t_1 X_1} e^{t_2 X_2} \right)$

$\Rightarrow t_2 = 0 \quad M_{X_1}(t_1) = E\left(e^{t_1 X_1} \right)$

混合矩: $\left. \dfrac{\partial^{\sum_{i=1}^{n} \alpha_i} M}{(\partial t_1)^{\alpha_1} \cdots (\partial t_n)^{\alpha_n}} \right|_{(0, 0, \cdots, 0)} = E\left(\prod_{i=1}^{n} X_i^{\alpha_i} \right)$

$Cov(X_1, X_2) = \dfrac{\partial^2 M}{\partial t_1 \partial t_2}(0, 0 \cdots 0) - \dfrac{\partial M}{\partial t_1}(0, \cdots 0) \dfrac{\partial M}{\partial t_2}(0, \cdots 0)$

相互独立 $\Leftrightarrow M(t_1, \cdots t_n) = M_{X_1}(t_1) \cdots M_{X_n}(t_n)$

例 1. $X, Y \overset{iid}{\sim} N(0, 1) \Rightarrow X+Y \perp X-Y$

Proof: $M_{X+Y, X-Y}(t_1, t_2) = E\left(e^{(X+Y) t_1} e^{(X-Y) t_2} \right)$

$= E\left(e^{X(t_1+t_2)} \right) E\left(e^{Y(t_1-t_2)} \right)$

$= e^{\frac{1}{2}[t_1^2 + t_2^2 + 2t_1 t_2]} e^{\frac{1}{2}(t_1^2 + t_2^2 - 2t_1 t_2)}$

$= e^{t_1^2 + t_2^2}$

另 $M_{X+Y}(t_1) \overset{t_2 = 0}{=\!=} E\left(e^{(X+Y) t_1} \right) = e^{\frac{1}{2} t_1^2} e^{\frac{1}{2} t_1^2} = e^{t_1^2}$ ✓

于是 $M_{X+Y, X-Y}(t_1, t_2) = M_{X+Y}(t_1) M_{X-Y}(t_2)$ #

例2. $X \perp Y \sim N(0,1)$ 固定 $\theta \in [0, 2\pi)$. $W = X\cos\theta - Y\sin\theta$

$Z = X\sin\theta + Y\cos\theta$

$\Rightarrow W, Z \overset{iid}{\sim} N(0,1)$

proof: $M_{W,Z}(t_1,t_2) = E\left(e^{t_1(X\cos\theta - Y\sin\theta)} e^{t_2(X\sin\theta + Y\cos\theta)} \right)$

$= E\left(e^{X(t_1\cos\theta + t_2\sin\theta)} e^{Y(-t_1\sin\theta + t_2\cos\theta)} \right)$

$= e^{\frac{1}{2}(t_1^2c^2 + t_2^2s^2 + 2t_1t_2sc) + \frac{1}{2}(t_1^2s^2 + t_2^2c^2 - 2t_1t_2sc)}$

$= e^{\frac{1}{2}(t_1^2 + t_2^2)} = M_W(t_1) M_Z(t_2)$ #.

例 Poisson especial.

例: X_1 表示记录 "p"

X_2 未记录 "1-p"

$X = X_1 + X_2 \sim P(\lambda)$ 等人数.

在 $X = n$ 条件下 $X_1 \sim B(n,p)$

$X_2 \sim B(n, 1-p)$.

$X_1 \sim P(\lambda p)$

$E\left(e^{(t_1 - t_2)X_1} \middle| X \right)$

$= \left(q + p e^{(t_1 - t_2)} \right)^X = \exp\left(X \ln(q + p e^{t_1 - t_2}) \right)$

$M_{X_1,X_2}(t_1,t_2) = E\left(e^{t_1X_1} e^{t_2X_2} \right)$

$= E\left(e^{t_2X_2} e^{(t_1 - t_2)X_1} e^{t_2X_1} \right)$

$= E\left(e^{t_2X} e^{(t_1 - t_2)X_1} \right)$

全期望 $E\left(E(e^{t_2X} e^{(t_1 - t_2)X_1} \middle| X) \right)$

$= E\left(e^{t_2X} E\left(e^{(t_1 - t_2)X_1} \middle| X \right) \right)$

$\Rightarrow E\left(e^{t_2X} e^{X\ln(q + p e^{(t_1 - t_2)})} \right)$

$= E\left(e^{\left[t_2 + \ln(q + p e^{t_1 - t_2}) \right] X} \right)$

$= e^{\lambda\left[e^{t_2 + \ln(q + p e^{t_1 - t_2})} - 1 \right]}$

$= e^{\lambda p(e^{t_1} - 1)} e^{\lambda q(e^{t_2} - 1)}$

$\Rightarrow X_1 \sim P(p\lambda)$